The IoT envisions a self-configuring, adaptive, complex network that interconnects 'things' to the Internet by using standard communication protocols. The interconnected things have physical or virtual representation in the digital world, sensing/actuation capabilities, a programmability feature and a unique identity. Gartner predicts that by 2020, 95 per cent of all new products will use IoT technology.

IoT platforms: An introduction

IoT platforms are regarded as the most critical component of the IoT ecosystem. Any IoT device has to connect to other IoT devices and applications (mostly cloud-based) to transfer information using standard Internet protocols. The gap between the device sensors and data networks is filled by IoT platforms. IoT platforms connect the data to the sensor arrangement, and provide insights using back-end applications to make sense of the plethora of data generated by the countless sensors.

A complete IoT system requires the following:

- Hardware such as sensors or devices. The task of sensors is to acquire data from the environment and perform actions accordingly.
- Connectivity in order to transmit the data to the cloud and a way to receive commands from the cloud. For some IoT systems, this bridge between the hardware and the cloud could be a router or a gateway in the network.
- Software that is installed on the cloud, the main task of which is to analyse the data collected from sensors and perform the task of decision making.
- A user interface, to facilitate user interaction. This can be a Web interface, where collected data is represented as numbers or graphs, or it could be a mobile app installed on a smartphone to interpret data and issue alerts to the end users, depending on sensor environments.

In a nutshell, IoT platforms support software that facilitates connectivity between IoT systems. These platforms facilitate communication, data flow, device management and app functionality.

Types of IoT platforms

There are, essentially, four types of IoT platforms.

- ***End-to-end IoT platforms*** facilitate the handling of millions of simultaneous device connections by providing hardware, software, connectivity, security and device management tools. In addition, these platforms provide OTA (over the air) firmware updates, cloud connectivity and device management to facilitate device monitoring.
- ***Connectivity platforms*** provide cheap and low power connectivity solutions via Wi-Fi and cellular technology.
- ***Cloud platforms*** reduce the complexity of building complex network stacks and provide back-end facilities for device monitoring.
- ***Data platforms*** based on IoT platforms provide numerous tools for data routing, and facilitate management and visualisation of data using data analytics tools.

Evaluation of IoT platforms

In order to select the best IoT based platform, the following factors need to be considered.

- ***Scalability:*** The IoT platform should be scalable enough to accommodate growing needs without any hiccup. The platform should support dynamic growth, 24x7 uptime and efficient backup to prevent all sorts of errors.
- ***Reliability:*** The IoT platform should handle all sorts of failover and have disaster recovery options to keep the overall system up and running without any issues. It is very important for every end user to pay sufficient attention to the reliability of the platform, which involves a combination of parameters linked to the architecture and operations.
- ***Customisation:*** IoT platforms should support lots of customisations and be well integrated with cloud services. They should consist of APIs, and have extensive libraries as well as strong integration with other platforms to enhance the core functionality of the system with the programmer's customised code.
- ***Operations:*** The IoT platform's operations should not be hidden; rather the end user should be greeted with a centralised interface which gives all the details with regard to system statistics, hardware information and the modules or services.
- ***Protocols:*** The most important and foremost protocol to facilitate IoT is MQTT. The IoT platform should support the MQTT protocol to provide all sorts of communication from sensors to the cloud. Almost all cloud service providers use MQTT. These include Microsoft Azure, Amazon AWS, Google Cloud, etc. The IoT platform should support the API over WebSockets, REST and CoAP.
- ***Hardware support:*** To choose an effective IoT platform, it is very important that the hardware facilitates low-end devices by using MQTT or CoAP protocols and, overall, the systems should be flexible and powerful enough to run programs written in any language.
- ***Cloud technology:*** The IoT platform should be flexible enough to support all operations without any hiccups, whether in the cloud, on premise or hybrid.
- ***Technical support:*** IoT platforms should have strong back-end technical support, and service providers should ensure regular updation of the platform with all the basic upgrades, security patches and bug fixes. Strong support develops confidence among end users.
- ***Systems architecture:*** An IoT platform should deploy production-grade, well-supported frameworks, tools and languages, making the platform easy to operate and flexible enough to support all sorts of custom implementations.
- ***Security:*** This is another significant parameter for an IoT platform. The service provider should offer SSL/TLS based communication with devices and applications, along with authentication for devices and users. It is very